

Maria Teleki

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SUMMARY

I develop methods, benchmarks, and audits that enable AI systems — and the institutions that deploy them — to be accountable to the people they serve. My research spans speech AI, computational social science, and HCI, across three threads:

- **Disfluency-Aware AI:** *Does AI work for the person speaking to it?* (ICASSP '26, INTERSPEECH '25): Real human speech is full of disfluency — pauses, repairs, and restarts — that current systems often treat as noise. I develop benchmarks and evaluation frameworks that model these patterns as structured signal, revealing robustness failures in conversational AI systems operating on real speech.
- **Personalized AI:** *Does AI work for the person it represents?* (ICWSM '25, EMNLP Findings '25, IUI '26): AI systems respond differently to different people — shaped by signals of who we are, like gender, accent, and voice, and signals of how we express ourselves, like creativity and discourse. Understanding these signals is what it takes to build personalized AI that serves every user.
- **Governable AI:** *Does AI work for the public it answers to?* (FAccT '26): I study the infrastructures of power — governments, universities, and tech companies — that determine how AI systems are authorized, deployed, and contested. I develop audits and systems that reveal where accountability breaks down.

EDUCATION

2022 – Present | PhD Computer Science at Texas A&M University

Advised by James Caverlee.

2017 – 2022 | B.S. Computer Science at Texas A&M University

Summa Cum Laude. Emphasis in Electrical Engineering. Minor in Math.

EXPERIENCE

Google DeepMind | Student Researcher | Cambridge, MA | June – Sep '26

Joining Summer 2026 to work on Speech AI with Brian Eoff and Rajat Hebbar.

RetailMeNot | Software Engineering Intern | Austin, TX | May – August '21

BERT embeddings and PCA to surface which semantic dimensions of coupon content predicted CTR.

The Hi, How Are You Project | Volunteer | Austin, TX | May – Dec '20

Built the “Friendly Frog” Alexa Skill with the organization at the beginning of the COVID-19 pandemic to promote mental health.

RetailMeNot | Software Engineering Intern | Austin, TX | May – August '20

Built the “RetailMeNot DealFinder” Alexa Skill to help users activate cash back offers.

Silicon Labs | Applications Engineering Intern | Austin, TX | May – August '19

Built a Python library adopted by customers for real-time hardware bus traffic analysis across three protocols.

PUBLICATIONS

- **1ST AUTHOR** Maria Teleki, Anna Seo Gyeong Choi, Anne Duray, Haoran Liu, Julie Zhang, Xiangjue Dong, Dilma Da Silva, Allison Koenecke, and James Caverlee, *The Due Process Deficit: Auditing AI Governance in U.S. Higher Education*. **FAccT '26**. — Collab. with Cornell, AAAS Science and Technology Policy Fellow. Presented at IC2S2 (Oral) '26.

- **1ST AUTHOR** Maria Teleki, Sai Janjur, Haoran Liu, Oliver Grabner, Ketan Verma, Thomas Docog, Xiangjue Dong, Lingfeng Shi, Cong Wang, Stephanie Birkelbach, Jason Kim, Yin Zhang, and James Caverlee, *Z-Scores: A Metric for Linguistically Assessing Disfluency Removal*. **ICASSP '26**
- Xiangjue Dong, Cong Wang, Maria Teleki, Millennium Bismay, Ruihong Huang, and James Caverlee, *CHOIR: Collaborative Harmonization for Inference Robustness*. **ACL Main '26**
- Shree Harsha Bokkalahalli Satish, Christoph Minixhofer, Maria Teleki, James Caverlee, Ondrej Klejch, Peter Bell, Gustav Eje Henter, and Éva Székely, *The Voice Behind the Words: Quantifying Intersectional Bias in SpeechLLMs*. **INTERSPEECH '26**. — Collab. with KTH, University of Edinburgh
- Shree Harsha Bokkalahalli Satish, Maria Teleki, Christoph Minixhofer, Ondrej Klejch, Peter Bell, and Éva Székely, *Walk a Mile in My Voice: Voice Conversion Shapes Trust, Attribution, and Empathy in Human-AI Speech Interactions*. **IUI '26 (Short)**. — Collab. with KTH, University of Edinburgh. Presented at CHI '26 (Poster).
- **1ST AUTHOR** Maria Teleki, Lingfeng Shi, Chengkai Liu, and James Caverlee, *I want a horror – comedy – movie: Slips-of-the-Tongue Impact Conversational Recommender System Performance*. **INTERSPEECH '25**
- **1ST AUTHOR** Maria Teleki, Xiangjue Dong, Haoran Liu, and James Caverlee, *Masculine Defaults via Gendered Discourse in Podcasts and Large Language Models*. **ICWSM '25**. Presented at IC2S2 (Oral) '25 and SCon@ACL '25.
- **1ST AUTHOR** Maria Teleki, Vedangi Bengali, Xiangjue Dong, Sai Tejas Janjur, Haoran Liu, Tian Liu, Cong Wang, Ting Liu, Yin Zhang, Frank Shipman, and James Caverlee, *A Survey on LLMs for Story Generation*. **EMNLP Findings '25**
- **1ST AUTHOR** Maria Teleki, Xiangjue Dong, Soohwan Kim, and James Caverlee, *Comparing ASR Systems in the Context of Speech Disfluencies*. **INTERSPEECH '24**
- **1ST AUTHOR** Maria Teleki, Xiangjue Dong, and James Caverlee, *Quantifying the Impact of Disfluency on Spoken Content Summarization*. **LREC-COLING '24**
- Rohan Chaudhury, Maria Teleki, Xiangjue Dong, and James Caverlee, *DACL: Disfluency Augmented Curriculum Learning for Fluent Text Generation*. **LREC-COLING '24**
- Xiangjue Dong, Ziwei Zhu, Zhuoer Wang, Maria Teleki, and James Caverlee, *Co2PT: Mitigating Bias in Pre-trained Language Models through Counterfactual Contrastive Prompt Tuning*. **EMNLP Findings '23**. — Collab. with George Mason University
- **SYSTEM** Majid Alfifi, Xiangjue Dong, Timo Feldman, Allen Lin, Karthic Madanagopal, Aditya Pethe, Maria Teleki, Zhuoer Wang, Ziwei Zhu, and James Caverlee, *Howdy Y'all: An Alexa TaskBot*. **Alexa Prize TaskBot Challenge Proceedings '22**

UNDER SUBMISSION

- **1ST AUTHOR** Maria Teleki, Sai Janjur, Haoran Liu, Oliver Grabner, Ketan Verma, Thomas Docog, Xiangjue Dong, Lingfeng Shi, Cong Wang, Stephanie Birkelbach, Jason Kim, Yin Zhang, Éva Székely, and James Caverlee, *Conversational Speech Reveals Systematic Robustness Failures in LLMs*. **INTERSPEECH (Long)**. — Collab. with KTH
- **CO-1ST AUTHOR** Xiangjue Dong*, Maria Teleki*, and James Caverlee, *A Survey on LLM Inference-Time Self-Improvement*. **COLM**

IN PREPARATION

- **CO-1ST AUTHOR** Maria Teleki*, Maike Züfle*, Xiangjue Dong, Vilém Zouhar, Fabian Retkowski, Oliver Grabner, Alex Waibel, Jan Niehues, and James Caverlee, *Uh-Mazing: A Multilingual Disfluency-Aware Speech Translation Benchmark*. **TACL**. — Collab. with Texas A&M University, KIT, ETH Zürich

- **CO-1ST AUTHOR** Anna Seo Gyeong Choi*, Maria Teleki*, Miguel del Rio, James Caverlee, Corey Miller, and Allison Koenecke, *SpeechSpectrum: A Framework for Speech-to-Text Representation Along the Linguistic Fidelity Spectrum*. **CSCW**. — Collab. with Cornell, Rev AI. Presented at IC2S2 '26.
- Anna Seo Gyeong Choi, Maria Teleki, James Caverlee, Miguel del Rio, Corey Miller, and Hoon Choi, *Beyond Single Ground Truth: Reference Monism as Epistemic Injustice in ASR Evaluation*. **AIES**. — Collab. with Cornell, Rev AI, Kangwon National University. Presented at IC2S2 '26.
- **CO-1ST AUTHOR** **UG 1ST AUTHOR** Jason Kim*, Maria Teleki*, Jobin Varughese*, and James Caverlee, *PromptHelper: A Prompt Recommender System for Education*. **CHI**. Presented at IC2S2 '26.
- Kiyrah Mowry, You Wu, Alexander Gaidarov, Maria Teleki, and Meng Xia, *AnnoMath: Generating Multimodal Math Explanations with Graphical Annotations*. **CHI**

MENTEE PAPERS, POSTERS & DEMOS

- **POSTER** **UG 1ST AUTHOR** Jason Kim, Maria Teleki, and James Caverlee, *PromptHelper: A Prompt Recommender System for Encouraging Creativity in AI Chatbot Interactions*. **CHI '26 (Poster)**. Presented at IC2S2 '26.
- **SHORT** **UG 1ST AUTHOR** Stephanie Birkelbach, Maria Teleki, Nehul Bhatnagar, Peter Carragher, Xiangjue Dong, and James Caverlee, *SocialPulse: An Open-Source Subreddit Sensemaking Toolkit*. **SocialLLM @ ICWSM '26**. — Collab. with CMU, Revionics. Presented at IC2S2 (Oral) '26.
- **DEMO** **UG 1ST AUTHOR** Anna Irmetova, Haoran Liu, Maria Teleki, Peter Carragher, and James Caverlee, *Pod-Checker: An Interpretable Fact-Checking Companion for Podcasts*. **MisD @ ICWSM '26**. — Collab. with CMU
- **SHORT** **UG 1ST AUTHOR** Oluwadayo Bamgbelu, Maria Teleki*, Xiangjue Dong*, and James Caverlee, *Detecting and Mitigating Demographic Bias in LLM-Based Resume Evaluation*. **In Preparation for ARR (Short)**. Presented at IC2S2 '26.
- **MS 1ST AUTHOR** Junhyeon Kwon, Jobin Varughese, Maria Teleki, and Mohsen Dorodchi, *When Behavioral Traces Mislead: Evaluating Evidence Adequacy in Learning Analytics*. **In Preparation for AIES**. — Collab. with UNC Charlotte. Presented at IC2S2 (Oral) '26.
- **DEMO** Amanda Lacy, Maria Teleki, Esau Hutcherson, Jobin Varughese, Jun Kwon, Frank Shipman, and Tracy Hammond, *AI in the Loop: A Multimodal Assistant for Real-Time Lecture Comprehension*. **L@S Demo '26**

SERVICE

Workshop Organizer: Kimi V. Wenzel, Alisha Pradhan, Maria Teleki, Tobias Weinberg, Robin Netzorg, Alyssa Hillary Zisk, Anna Seo Gyeong Choi, Jingjin Li, Raja Kushalnagar, Colin Lea, Abraham Glasser, Christian Vogler, Nan Bernstein Ratner, Ly Xīnzhèn M. Zhǎngsūn Brown, Allison Koenecke, Karen Nakamura, and Shaomei Wu, *Speech AI for All: The What, How, and Who of Measurement*. **CHI '26 Workshop**

Program Committee (Reviewer) for **INTERSPEECH**: '26

Program Committee (Reviewer) for **IC2S2**: '26

Program Committee (Reviewer) for **ACL ARR**: Aug '24, Oct '24, Dec '24, May '25, Oct '25, Jan '26

Program Committee (Reviewer) for **ICWSM**: Jan '24, May '24, Sep '24, Jan '25, Sep '25, Jan '26

External Program Committee (External Reviewer) for **RecSys**: '24

TEACHING

Certifications

2025 | Professional Development Mastery Certificate in Instruction & Assessment | **GRAD Aggies, CIRTL@TAMU**
Completed multi-semester program in evidence-based teaching, including syllabus design, learning assessment, and inclusive pedagogy.

Teaching Assistantships

Spring 2026 | Teaching Assistant | Texas A&M University, CSCE 676 Data Mining & Analysis

Guest Lectures

Fall 2025 | Speech Language Models | Texas A&M University, CSCE 676 Data Mining & Analysis

Spring 2025 | IR Evaluation and Learning to Rank | Texas A&M University, CSCE 670 Information Storage & Retrieval

Earlier Teaching

Dec '18 – Dec '19 | Peer Teacher | Texas A&M University

Sep '16 – July '17 | Afterschool Instructor | The Y (YMCA)

Taught weekly classes at local elementary schools and authored robotics & science instruction manuals for the Williamson County YMCA Afterschool program.

MENTORING

★ student was an author on a published paper; ♠ student was an author on a Preprint paper; ▲ student completed their thesis; ♦ student received course credit (i.e. CSCE 485, CSCE 691); ♣ student had no publications prior to mentorship.

MS Students

Rohan Chaudhury [★▲] – *First Employment at Amazon*

Sai Tejas Janjur [★♠♣] – 🏆 *1st Place Award at Student Research Week; First Employment at NVIDIA*

Cong Wang [★♠♣] – *Presented at Texas NLP Symposium*

Junhyeon Kwon [♠♣]

Undergraduate Students

Soohwan Kim [★♦♣] – *First Employment at UPS*

Oliver Grabner [★♠♣] – *Presented at Texas NLP Symposium & Student Research Week; Intern at Samsung; Intern at Google*

Thomas Docog [★♠♣] – *Presented at Texas NLP Symposium; Intern at Ramirez*

Ketan Verma [★♠♣] – *Intern at Samsung; MS at Cornell Tech*

Stephanie Birkelbach [★♠♦♣] – *REU at Rochester Institute of Technology w/ Ashique KhudaBukhsh*

Jason Kim [★♠♣] – *Presented at Texas NLP Symposium; 🏆 1st Place Award at Student Research Week*

Suhani Saxena [♣]

Anna Irmetova [★♣♦] – *Presented at Student Research Week; Intern at Target*

Oluwadayo (Dayo) Bamgbelu [♣] – *Presented at Texas NLP Symposium; Intern at Meta*

AWARDS

2022–2026 | Dr. Dionel Avilés '53 and Dr. James Johnson '67 Fellowship in Computer Science and Engineering
Multi-year fellowship supporting doctoral research in Computer Science.

2024, 2025, 2026 | Department of Computer Science & Engineering Travel Grant

2024 | CRA-WP Grad Cohort for Women

Selected participant; supported with travel support.

2017–2021 | President's Endowed Scholarship

Multi-year fellowship supporting undergraduate studies.

2018 | Bertha & Samuel Martin Scholarship

INVITED TALKS

Spring 2026 | *Accountable AI* | IPT Global

Fall 2025 | *Conversational AI* | Revionics

Spring 2025 | *How does ChatGPT work? & My research!* | Texas A&M University, Club of Aggie Female Engineers (C.A.F.E.)

Fall 2024 | *The Other AI: An Intuitive Understanding of Artificial Intelligence* | Texas Tech University, School of Veterinary Medicine VBMA